PARISHRAM 2.0

CLASS 12TH 2026

Inverse Trigonometric Functions

MATHS | LECTURE-4

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TOPICS TO BE COVERED

Revise

1. 11th Ki Formulas } For 12th
2. Questions Based on Formulae's

Property I, II, III & IV

Yaad h??

Why Fear When Rolex Sir is Here



Class 11th Trigo Formula for 12th

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

Class 11th Trigo Formula for 12th

- $\sin 2\theta = 2 \sin\theta \cos\theta$ $\quad \sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$
- $\cos 2\theta = \cos^2\theta - \sin^2\theta$ $\quad = \frac{1-\tan^2\theta}{1+\tan^2\theta}$
  $= 2\cos^2\theta - 1$
  $= 1 - 2\sin^2\theta$
- $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$
- $1 + \cos x = 2\cos^2\frac{x}{2}$
- $1 - \cos x = 2\sin^2\frac{x}{2}$

QUESTION

Simplify $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$ where $-\frac{3\pi}{2} < x < \frac{\pi}{2}$

$$\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$$

$$\tan^{-1}\left[\frac{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}{\cos^2\frac{x}{2}+\sin^2\frac{x}{2}-2\sin\frac{x}{2}\cos\frac{x}{2}}\right]$$

$$\tan^{-1}\left[\frac{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)}{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2}\right]$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$\sin 2\theta = 2\sin\theta\cos\theta$

$a^2+b^2-2ab=(a-b)^2$

$$= \tan^{-1}\left[\frac{\cos\frac{x}{2}+\sin\frac{x}{2}}{\cos\frac{x}{2}-\sin\frac{x}{2}}\right]$$

Divide Nr & Dr by $\cos\frac{x}{2}$

$$= \tan^{-1}\left[\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}\right]$$

$$= \tan^{-1}\left[\frac{1+\tan\frac{x}{2}}{1-(1)\tan\frac{x}{2}}\right]$$

$$= \tan^{-1}\left[\frac{\tan\frac{\pi}{4}+\tan\frac{x}{2}}{1-\tan\frac{\pi}{4}\tan\frac{x}{2}}\right]$$

$$= \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right]$$

$$= \frac{\pi}{4}+\frac{x}{2}$$

$$-\frac{3\pi}{2\times2} < \frac{x}{2} < \frac{\pi}{2\times2}$$

$$-\frac{3\pi}{4} < \frac{x}{2} < \frac{\pi}{4}$$

$$-\frac{3\pi}{4}+\frac{\pi}{4} < \frac{x}{2}+\frac{\pi}{4} < \frac{\pi}{4}+\frac{\pi}{4}$$

$$-\frac{2\pi}{4} < \frac{x}{2}+\frac{\pi}{4} < \frac{2\pi}{4}$$

$$-\frac{\pi}{2} < \frac{x}{2}+\frac{\pi}{4} < \frac{\pi}{2}$$

$$1 = \tan\frac{\pi}{4}$$

$$\tan(A+B)$$

$$\rightarrow \frac{\tan A+\tan B}{1-\tan A\tan B}$$

QUESTION

Express $\sin^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right)$ where $-\frac{\pi}{4} < x < \frac{\pi}{4}$, in the simplest form.

$$\sin^{-1}\left(\sin x\left(\frac{1}{\sqrt{2}}\right) + \cos x\left(\frac{1}{\sqrt{2}}\right)\right)$$

$$\sin^{-1}\left(\sin x \times \cos\frac{\pi}{4} + \cos x \times \sin\frac{\pi}{4}\right)$$

$$\sin A\cos B + \cos A \sin B = \sin(A+B)$$

$$= \sin^{-1}\left(\sin\left(x + \frac{\pi}{4}\right)\right)$$

$$= x + \frac{\pi}{4}$$

$$-\frac{\pi}{4} < x < \frac{\pi}{4}$$

$$-\frac{\pi}{4} + \frac{\pi}{4} < x + \frac{\pi}{4} < \frac{\pi}{4} + \frac{\pi}{4}$$

$$0 < x + \frac{\pi}{4} < \frac{\pi}{2}$$

QUESTION

$\sin x = \frac{2\sqrt{2}}{3}$

$\hookrightarrow x = \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

$\cos^{-1}\left(\frac{1}{3}\right) = x$

$\frac{1}{3} = \cos x$

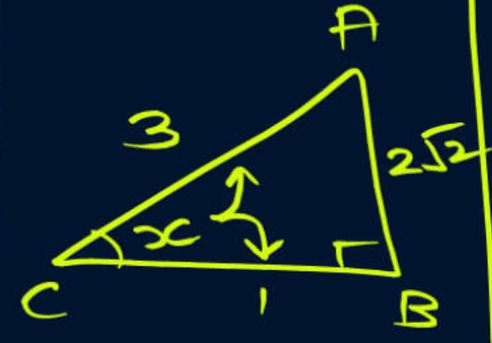
Prove that : $\frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right) = \frac{9}{4}\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

LHS: $\frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right)$

$\frac{9}{4}\left(\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{3}\right)\right)$

$\frac{9}{4}\left(\cos^{-1}\left(\frac{1}{3}\right)\right)$

$\frac{9}{4}(x) = \frac{9}{4}\left(\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)\right)$



$3^2 = 1^2 + AB^2$

$9 = 1 + AB^2$

$8 = AB^2$

$\sqrt{8} = AB$

$\boxed{AB = 2\sqrt{2}}$

$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

$\sin^{-1}\left(\frac{1}{3}\right) + \cos^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{2}$

$\cos^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{3}\right)$

QUESTION

What would be the value of $\cos^{-1}\left(\frac{24}{25}\right) + \tan^{-1}\left(\frac{24}{7}\right)$?

y

$y + \tan^{-1}\left(\frac{24}{7}\right)$

$= \cot^{-1}\left(\frac{24}{7}\right) + \tan^{-1}\left(\frac{24}{7}\right)$

$= \tan^{-1}\left(\frac{24}{7}\right) + \cot^{-1}\left(\frac{24}{7}\right)$

$= \frac{\pi}{2}$

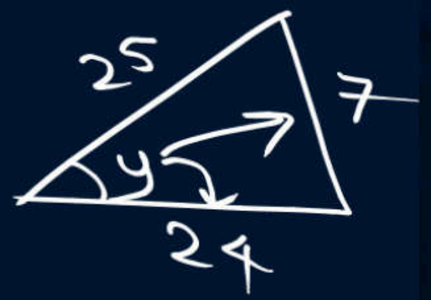
$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$

$y = \cos^{-1}\left(\frac{24}{25}\right)$

$\cos y = \frac{24}{25}$

$\sin y = \frac{7}{25}$

$\cot y = \frac{24}{7}$



$y = \cot^{-1}\left(\frac{24}{7}\right)$

QUESTION

Find the value of $\sin\left(\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{5}{12}\right)\right)$

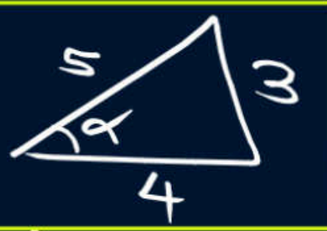
α $\quad$ β

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

$$\sin(\alpha+\beta)$$
$$=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$
$$=\frac{3}{5}\times\frac{12}{13}+\frac{4}{5}\times\frac{5}{13}$$
$$=\frac{36}{65}+\frac{20}{65}=\frac{36+20}{65}=\frac{56}{65}$$

$\alpha=\cos^{-1}\left(\frac{4}{5}\right)$

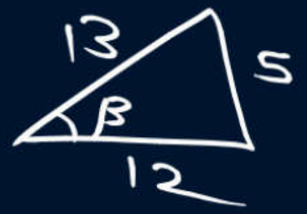
$\cos\alpha=\frac{4}{5}$



$\sin\alpha=\frac{3}{5}$

$\beta=\tan^{-1}\left(\frac{5}{12}\right)$

$\tan\beta=\frac{5}{12}$



$\sin\beta=\frac{5}{13}$

$\cos\beta=\frac{12}{13}$

QUESTION

Prove that : $\sin^{-1}\left(\frac{63}{65}\right) = \sin^{-1}\left(\frac{5}{13}\right) + \cos^{-1}\left(\frac{3}{5}\right)$

α $\qquad$ β

RHS $= \alpha + \beta = \sin^{-1}\left(\frac{63}{65}\right)$

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

$= \frac{5}{13} \times \frac{3}{5} + \frac{12}{13} \times \frac{4}{5}$

$= \frac{15}{65} + \frac{48}{65}$

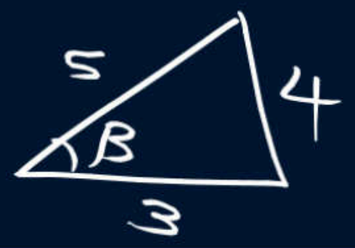
$= \frac{15+48}{65} = \frac{63}{65}$

$\sin(\alpha+\beta) = \frac{63}{65}$

$(\alpha+\beta) = \sin^{-1}\left(\frac{63}{65}\right)$

$\beta = \cos^{-1}\left(\frac{3}{5}\right)$

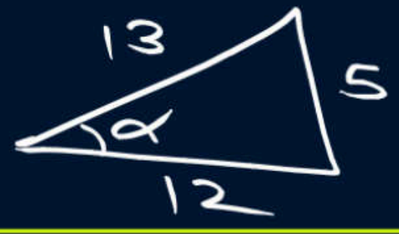
$\cos\beta = \frac{3}{5}$



$\sin\beta = \frac{4}{5}$

$\alpha = \sin^{-1}\left(\frac{5}{13}\right)$

$\sin\alpha = \frac{5}{13}$



$\cos\alpha = \frac{12}{13}$

QUESTION

Prove that : $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\frac{a}{b}\right)=\frac{2b}{a}$

$\frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right)=x$

$\tan\left(\frac{\pi}{4}+x\right)+\left(\tan\frac{\pi}{4}-x\right)$

$=\frac{\tan\frac{\pi}{4}+\tan x}{1-\tan\frac{\pi}{4}\tan x}+\frac{\tan\frac{\pi}{4}-\tan x}{1+\tan\frac{\pi}{4}\tan x}$

$=\frac{1+\tan x}{1-\tan x}+\frac{1-\tan x}{1+\tan x}$

$=\frac{(1+\tan x)^2+(1-\tan x)^2}{(1-\tan x)(1+\tan x)}$

$=\frac{1+2\tan x+\tan^2 x+1-2\tan x+\tan^2 x}{(1-\tan x)(1+\tan x)}$

$=\frac{2+2\tan^2 x}{(1-\tan x)(1+\tan x)}$

$=\frac{2(1+\tan^2 x)}{(1-\tan^2 x)}$

$$\frac{2(1+\tan^2 x)}{(1-\tan^2 x)}$$

$$= \frac{2}{\boxed{\frac{(1-\tan^2 x)}{(1+\tan^2 x)}}}$$

$$= \frac{2}{\cos 2x}$$

$$= \frac{2}{\cos 2\left(\frac{1}{2}\cos^{-1}\frac{a}{b}\right)}$$

$$x = \frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right) \rightarrow \text{Start assume}$$

$$= \frac{2}{\cos\left(\cos^{-1}\left(\frac{a}{b}\right)\right)}$$

$$= \frac{2}{\frac{a}{b}}$$

$$= \frac{2b}{a}$$

LHS = RHS

Hence Proved

Summary

Parishram

Lots of Love

from Mandar

THANK YOU